Convolution of a Sinusoidal Signal with a Shifted Rectangular Kernel

Group Project Report

April 27, 2025

1 Problem Statement

In this analysis, we investigate the convolution of a sinusoidal input signal $f(t) = A\sin(\omega t + \phi)$ with a rectangular kernel that is shifted by τ_0. The shifted rectangular kernel is defined as:

$$h(t) = \begin{cases} 1, & \text{for } \tau_0 - T < t < \tau_0 + T \\ 0, & \text{otherwise} \end{cases}$$

Our goal is to derive the convolution expression $y(t) = (f * h)(t)$ analytically and analyze the system's behavior.

2 Analytical Derivation

The convolution of $f(t)$ and $h(t)$ is defined as:

$$y(t) = (f * h)(t) = \int_{-\infty}^{\infty} f(\tau)h(t-\tau)d\tau$$

Substituting our functions:

$$y(t) = \int_{-\infty}^{\infty} A\sin(\omega\tau + \phi) \cdot h(t-\tau)d\tau$$

Since the kernel $h(t-\tau)$ is non-zero only when $\tau_0 - T < t - \tau < \tau_0 + T$, we can rewrite this as:

$$t - \tau_0 - T < \tau < t - \tau_0 + T$$

Therefore, the limits of integration become:

$$y(t) = \int_{t-\tau_0-T}^{t-\tau_0+T} A\sin(\omega\tau + \phi)d\tau$$

Evaluating this integral:

$$
\begin{aligned}
y(t) &= A \int_{t-\tau_0-T}^{t-\tau_0+T} \sin(\omega\tau + \phi) d\tau \\
&= -\frac{A}{\omega} [\cos(\omega\tau + \phi)]_{t-\tau_0-T}^{t-\tau_0+T} \\
&= -\frac{A}{\omega} [\cos(\omega(t - \tau_0 + T) + \phi) - \cos(\omega(t - \tau_0 - T) + \phi)]
\end{aligned}
$$

Using the trigonometric identity $\cos(\alpha) - \cos(\beta) = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$:

$$
y(t) = -\frac{A}{\omega}\left[-2\sin\left(\frac{\omega(t-\tau_0+T)+\phi+\omega(t-\tau_0-T)+\phi}{2}\right)\right] \left[\sin\left(\frac{\omega(t-\tau_0+T)+\phi-(\omega(t-\tau_0-T)+\phi)}{2}\right)\right]
$$

$$
\begin{aligned}
y(t) &= -\frac{A}{\omega}\left[-2\sin(\omega(t-\tau_0)+\phi)\sin(\omega T)\right] \\
\implies y(t) &= \frac{2A\sin(\omega T)}{\omega}\sin(\omega(t-\tau_0)+\phi)
\end{aligned}
$$

We can see that this result is similar to the case where shift does not happen. The main difference which is seen that the convolution output is shifted by an exact amounnt τ_0.

3 Analysis of the Result

The convolution result can be expressed as:

$$
y(t) = \frac{2A\sin(\omega T)}{\omega}\sin(\omega(t-\tau_0)+\phi)
$$

This shows that:

1. The output is still a sinusoidal function with the same frequency ω as the input.
2. The amplitude is scaled by a factor of $\frac{2\sin(\omega T)}{\omega}$, which depends on both the frequency of the input signal and the half-width T of the rectangular kernel.
3. The phase of the output signal is shifted by τ_0, which corresponds to the shift in the rectangular kernel.

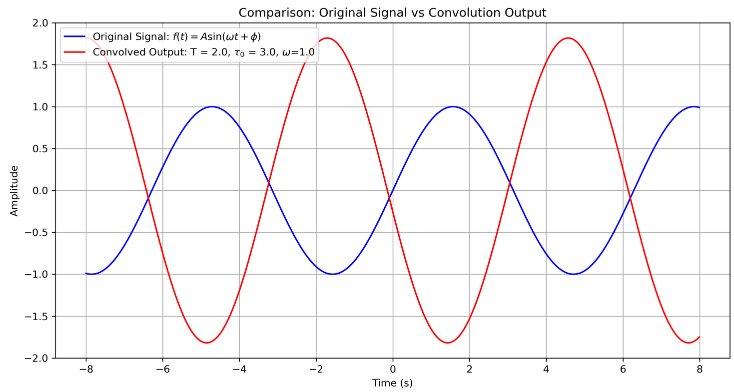


Figure 1: Original Signal vs Convolved Signal plot

4 Effect of Kernel Shift τ_0

The shift τ_0 in the kernel results in a time delay in the output signal. This is evident from the term $\sin(\omega(t - \tau_0) + \phi)$ in our result. This demonstrates an important property of convolution: a shift in one of the functions results in an equivalent shift in the convolution result.

In the context of time-delayed systems, this shows that the system introduces a delay of τ_0 in the response. This could represent physical phenomena like signal propagation delays in communication systems or processing delays in control systems.

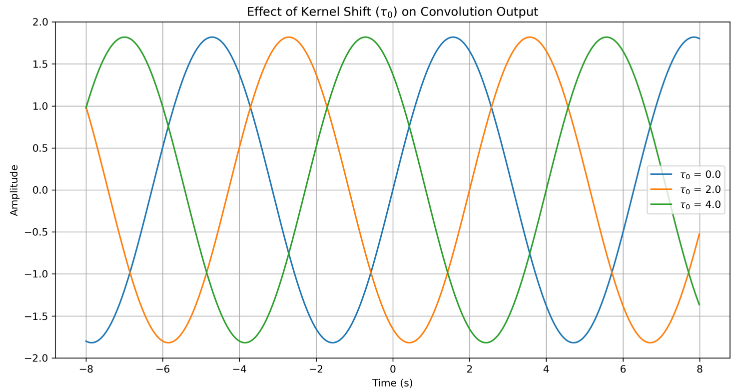
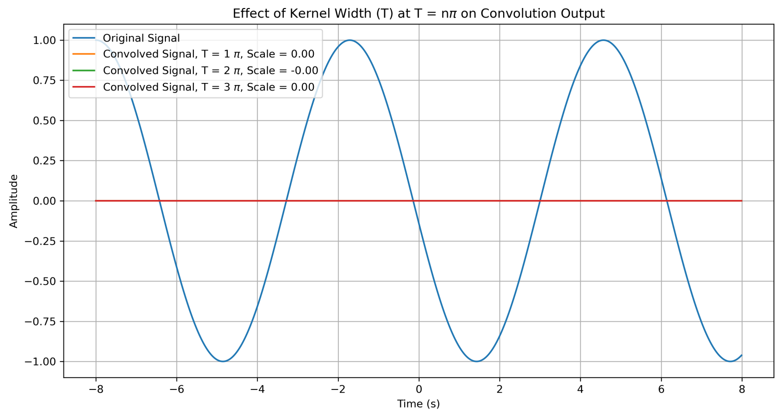


Figure 2: Effect of τ_0 on the convoluted signal

5 Special Cases

1. When $\omega T = n\pi$ (where n is a non-zero integer), $\sin(\omega T) = 0$, making the output zero. This represents frequencies that are completely filtered out by the system.



2. When T approaches zero, the kernel approaches a unit impulse function, and the output approaches zero for all t.

3. When T is very large, the term $\frac{\sin(\omega T)}{\omega}$ oscillates, but its amplitude decreases as $\frac{1}{\omega}$, acting as a low-pass filter.

6 Amplitude ratio Analysis

Amplitude ratio is given by:

$$\frac{A_{out}}{A_{in}} = \frac{2\sin\omega T}{\omega} = 2T sinc(\omega T) \tag{1}$$

As you can observe, it is a typical *sinc* function which dies down as ω increases. This makes it an interesting way to filter out high frequencies, effectively making it a working but crude low-pass filter.

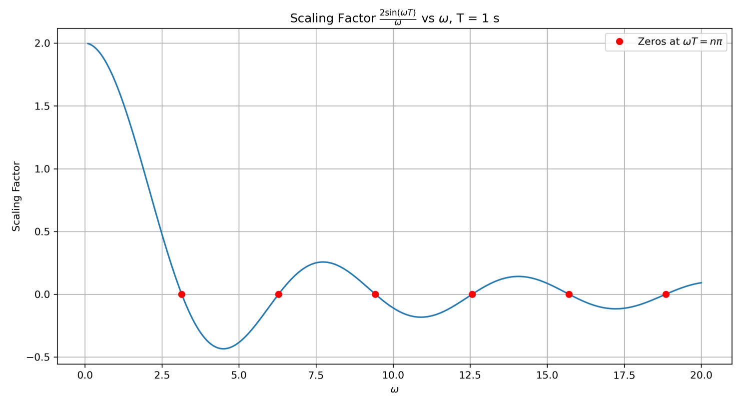


Figure 3: Amplitude/Scaling factor dependence on ω taking $T = 1s$

The amplitude ratio is sinusoidal with respect to T with angular frequency ω when all the other parameters are constant.

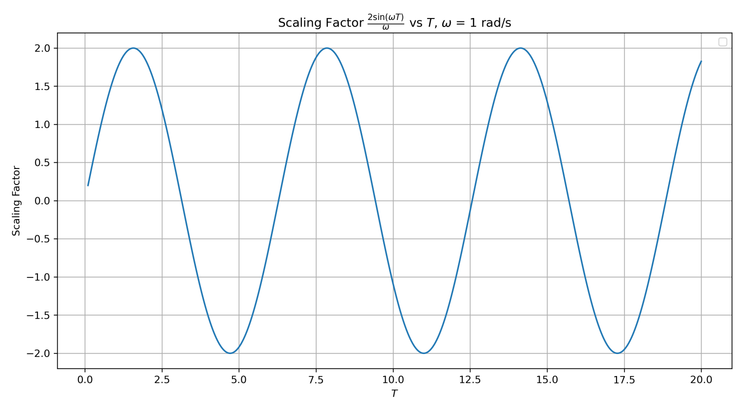


Figure 4: Amplitude/Scaling factor dependence on T taking $w = 1rad/s$

7 More analysis on Effect of Kernel Width (T)

The convolved ouput's amplitude ratio is sinusoidal with respect to T when all the other parameters are constant.

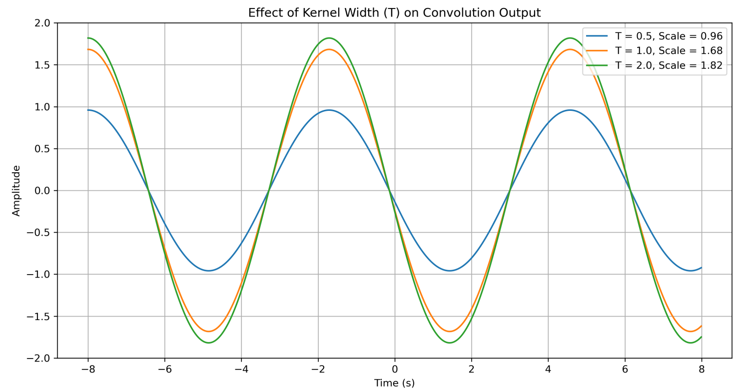


Figure 5: Effect of Varying T on the Convolved output

8 Smoothening of curves using the Box Kernel

When smoothing a signal $f(t)$ using the box kernel $h(t)$, we compute the convolution:

$$y(t) = (f * h)(t) = \int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau$$

For the rectangular kernel, this simplifies to:

$$y(t) = \int_{t-T}^{t+T} f(\tau)d\tau$$

This operation calculates the average value of $f(t)$ over a window of width $2T$ centered at point t.

This notion of "taking average" effectively "smoothens" or removes the noise of the curve it is convolved with. The box kernel acts as a low-pass filter, averaging out high-frequency noise while preserving the general shape of the signal. The choice of the kernel width T is crucial; a larger T results in greater smoothing but can distort finer features of the signal.

8.1 Properties of Box Kernel Smoothing

8.1.1 Effect of Kernel Width

The parameter T controls the degree of smoothing:

- Small T: Preserves more details but retains more noise

- Large T: Creates smoother curves but may obscure important features

8.1.2 Limitations

The box kernel produces a less smooth result compared to other kernels like Gaussian because of its abrupt cutoff at the boundaries. This creates a "choppy" appearance, especially in areas with sparse data.

8.2 Some noisy curves and their convolved output

Gaussian noise with mean 0 and standard deviation 0.02 is added to the original curve and then it is convolved with the box kernel.

8.2.1 Noisy and Smoothed Sinusoid

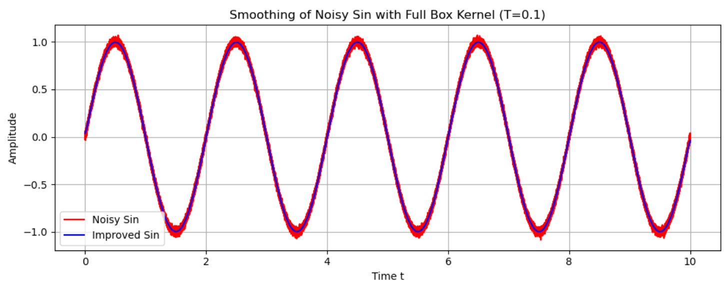


Figure 6: Noisy Sinusoid and Smoothed Sinusoid

8.2.2 Noisy and Smoothed Step Function

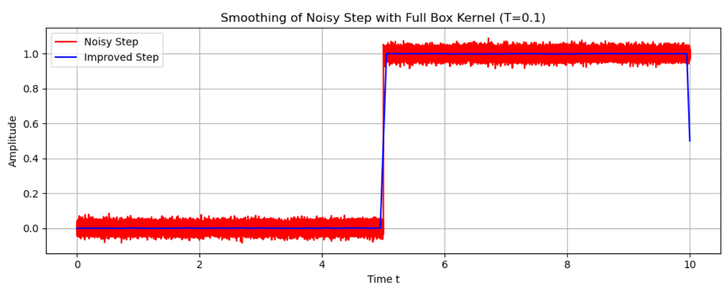


Figure 7: Noisy Step and Smoothed Step

8.2.3 Noisy and Smoothed Square Wave

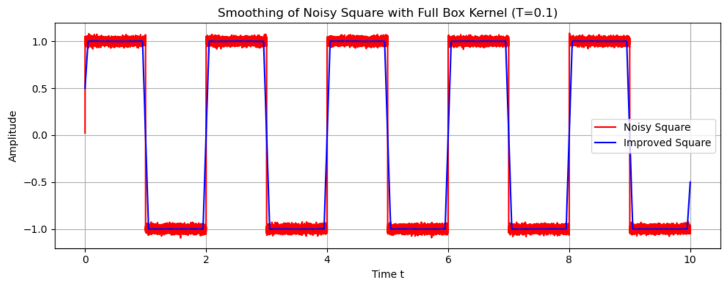


Figure 8: Noisy Square Wave and Smoothed Square Wave

8.2.4 Noisy and Smoothed Ramp

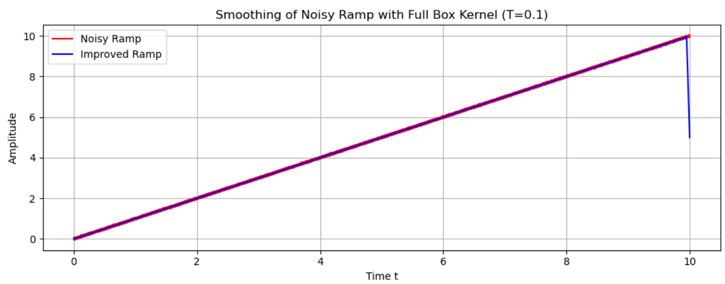


Figure 9: Noisy Ramp and Smoothed Ramp

8.2.5 Noisy and Smoothed Exponential Decay

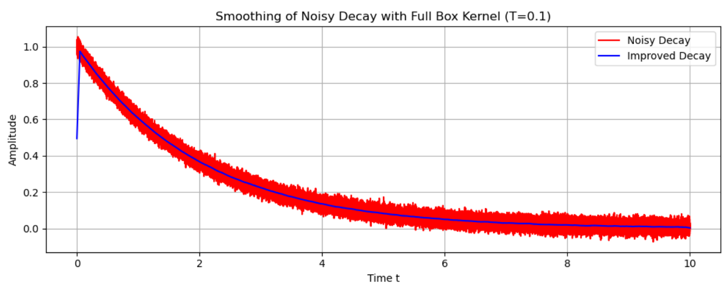


Figure 10: Noisy Exponential Decay and Smoothed Exponential Decay

8.2.6 Noisy and Smoothed Gaussian Curve

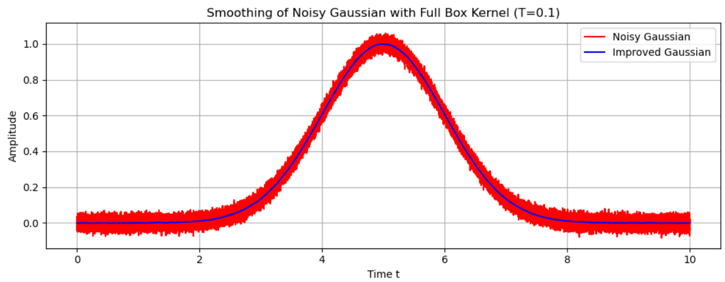


Figure 11: Noisy Gaussian and Smoothed Gaussian

9 Smoothening out Gibbs phenomena in Fourier Reconstruction using Box Kernel

Box kernel, in essence, smoothenes out noise. But an relatively interesting usecase would be to reduce the Gibbs Phenomena in the case of Fourier series.

9.1 Square Wave Fourier Coefficients

The square wave serves as an excellent example for demonstrating the Gibbs phenomenon. For a square wave alternating between $+1$ and -1 with period 2π, the Fourier series is:

$$f(t) = \frac{4}{\pi} \sum_{n=1,3,5,\ldots}^{\infty} \frac{1}{n} \sin(nt) \tag{2}$$

9.2 Convolving the square wave

By linearity of the convolution operator,

$$(f * h)(t) = \frac{4}{\pi} \sum_{n=1,3,5,\ldots}^{\infty} \left(\frac{2 \sin nT}{n} \right) \frac{1}{n} \sin(nt) \tag{3}$$

This alters the amplitude of the coefficients, which is obviously underisable. If we normalize this convolution amplitude factor, we get:

$$(f * h)(t) = \frac{4}{\pi} \sum_{n=1,3,5,\ldots}^{\infty} \left(\frac{\sin nT}{nT} \right) \frac{1}{n} \sin(nt) \tag{4}$$

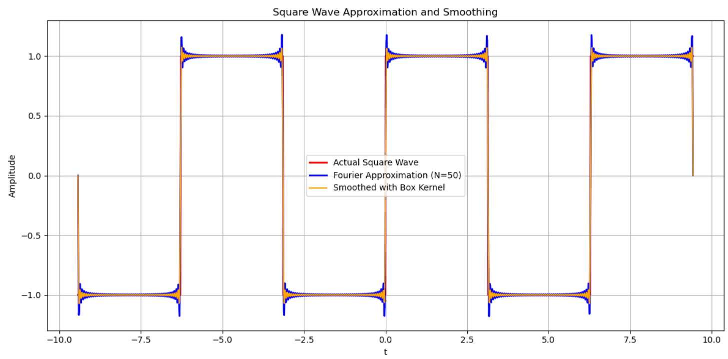


Figure 12: Slightly less Gibbs Phenomena in the Fourier Reconstruction at N = 50

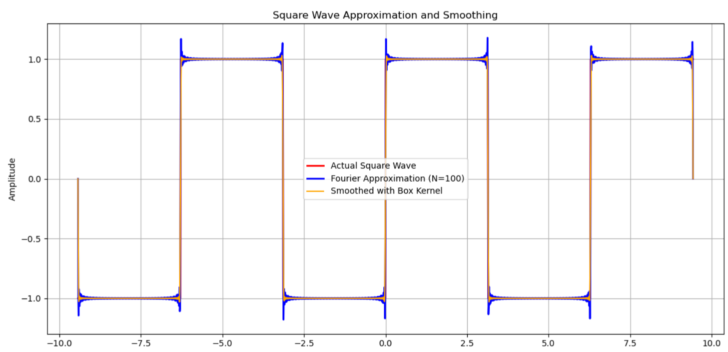


Figure 13: Slightly less Gibbs Phenomena in the Fourier Reconstruction at N = 100